MIL-C-48156(PA)

1 September 1974

MILITARY SPECIFICATION
CARTRIDGE, IGNITION, XM299
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing of the XM299, Ignition Cartridge for 81mm, HE, mortar ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard
L-T-90 - Tape, Pressure Sensitive

MILITARY

MIL-P-223 - Powder, Black
MIL-P-46425 - Primer, Percussion, M35, Loading Assembling and Packing
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-286 - Propellants, Solid, Sampling, Examination and Testing.

FSC: 1315

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DRAWINGS

PRODUCT AND PACKAGE DRAWINGS

9293422 - Cartridge, Ignition, XM299, Loading Assembly
9287505 - Box Fiberboard
9287506 - Box, Wirebound
9287507 - Tray assembly for Cartridge, Ignition XM299
9287508 - Packing, and Marking for Box Wirebound, Packing for Cartridge Ignition - XM299.
9278622 - Envelope Drawing for Leak Testing of Ignition Cartridges

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

49 CFR 71-90 Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles.

(Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite "49 CFR 71-90 (latest revision)."

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The ignition cartridge assembly shall comply with all requirements specified on Drawing (dwg) 9293422 and with all requirements specified in applicable specifications.

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3.3 Moisture Content of Black Powder Pellet.-The moisture content of the black powder pellet (Dwg. 7549177) at the time of assembling into the ignition cartridge shall not exceed 0.50 percent. Testing shall be as specified in 4.4.3.1.

3.4 Sensitivity

3.4.1 Non-functioning.-The primer shall not function under an applied energy value of 8.85 inch ounces minimum (min.) when tested as specified in 4.4.3.4.1.

3.4.2 Functioning.-The primer shall function under an applied energy value of 82.60 inch ounces maximum (max.) when tested as specified in 4.4.3.4.2.

3.5 Leak Test.-The loaded cartridges shall satisfactorily pass the leak test specified in 4.4.3.5.

3.6 Proving Ground Test.-The ignition cartridges shall comply with the following requirements (see 4.4.3.6).

3.6.1 Misfires.-The ignition cartridges shall function on the first attempt, without any audible hangfire.

3.6.2 Metal Part Security.-There shall be no evidence of rupture or metal parts separation in the gun bore or in flight.

3.6.3 Velocity and functioning (Proving ground).-The cartridge assemblies shall function and shall impart a corrected muzzle velocity of not more than 225 feet per second (fps) and not less than 205 fps. In addition, the velocity standard deviation of the lot shall be no greater than 2.4 fps.

3.7 Compatibility of Tape at 90°C-The reactivity value shall not exceed 5 ml of gas when determined as specified in 4.4.3.8.

3.8 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.9 Propellant charge weight assessment.-The M9 propellant lots shall be assessed to achieve a velocity of 215 fps. This will determine the weight of propellant to be loaded into each ignition cartridge. (see 4.5.6).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection.-The following types of inspection shall be conducted on this item:

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. Fifty (50) ignition cartridge assemblies shall be submitted for testing to a Government Proving Ground designated by the contracting officer. In addition, one hundred (100) assemblies shall be submitted for examination and tests at Picatinny Arsenal, Dover, New Jersey. All tests shall be performed with the equipment, and in accordance with the methods, specified in 4.5.

4.3.2 Inspections to be Performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspections and tests listed in paragraphs 4.4.2 through 4.4.3.

4.3.3 Rejection.-See MIL-A-48078(PA).

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of ignition cartridges shall contain:

a. Metal parts of one lot interfix from one manufacturer.
b. Black powder from not more than one lot from one manufacturer.

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c. Percussion elements from not more than one lot interfix from one manufacturer.

d. M9 propellant from not more than one lot from one manufacturer.

e. Tape from not more than one manufacturer or manufacturer part number.

f. Adhesive sealant from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9293423
4.3.2.1	Flash Tube Assembly				NEXT HIGHER ASSEMBLY 9293422
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	1. Flash holes not completely covered with tape		100%	3.2	Visual
	2. Pin missing or improper		100%	3.2	Visual
Major A	None defined				
Major B	None defined				
Minor					
201	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9285390
4.3.2.2	Tube Assembly		NEXT HIGHER ASSEMBLY 9293422

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Outside diameter, max		0.40%	3.2	Gage
132	Inside diameter, min		0.40%	3.2	Gage
133	Length of tape fold-in, max		0.40%	3.2	Gage
134	Length of tape overlap, max		0.40%	3.2	Gage
135	Complete coverage by tape without wrinkles or separation		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.3.2.3	Cartridge, Ignition, Prior to Assembly of Cap				9293422
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	1. Firing plug not loose after assembly		100%	3.2	Manual
	2. Flash tube assembly missing		100%	3.2	Visual
Major A					
101	Propellant missing		100%	3.2	Visual
102	Percussion element inverted		100%	3.2	Visual
103	Pellet missing		100%	3.2	Visual
Major B					
131	Presence of liner		100%	3.2	Visual
132	Overall length of body with primer		100%	3.2	Gage
133	Lacquer seal missing or inadequate around percussion element		0.40%	3.2	Visual
134	Adhesive sealant not applied to body groove for a full 360 degrees prior to assembling tube assembly		0.40%	3.2	Visual
135	Tube assembly not properly seated on body		0.40%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.3.2.3	Cartridge, Ignition, Prior to Assembly of Cap				9293422
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
136	Evidence of body and head not completely assembled.		0.40%	3.2	Visual
Minor					
201	Foreign matter in charge		0.65%	3.2	Visual
202	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.4	Cartridge, Ignition, After Assembly of Cap				9293422
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	RTV sealant missing, inadequate or excessive prior to or after assembly of cap		0.40%	3.2	Visual
132	Cap cocked or not completely seated		0.40%	3.2	Visual
133	Stake missing		0.40%	3.2	Visual
134	Length from shoulder to end, max		0.40%	3.2	Gage
135	Outside diameter, max.		0.40%	3.2	Gage
136	Concentricity		0.40%	3.2	Gage
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.5	Tray Assembly for Cartridge, Ignition XM299		9287507
			NEXT HIGHER ASSEMBLY 9287505

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Cartridges missing		0.40%	3.2	Visual
132	Tray broken, split or incomplete		0.40%	3.2	Visual
133	Improperly packed		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287505
4.3.2.6	Backed Box, Fiberboard				NEXT HIGHER ASSEMBLY 9287506
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Box broken or split		0.40%	3.2	Visual
132	Incomplete pack		0.40%	3.2	Visual
133	Improperly packed		0.40%	3.2	Visual
134	Improperly sealed		0.40%	3.2	Visual
Minor					
201	Marking incomplete or incorrect		0.65%	3.2	Visual
202	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.7	Box, Wirebound, Before Packing		9287506
			NEXT HIGHER ASSEMBLY 9287508

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Inside dimensions incorrect		0.40%	3.2	Gage
132	Broken, split or missing components		0.40%	3.2	Visual
Minor					
201	Marking incomplete or incorrect		0.65%	3.2	Visual
202	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.8	Packed Wirebound Box, Before Closing				9287508
					NEXT HIGHER ASSEMBLY -
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Improperly packed		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.9	Packed Wirebound Box, After Closing		9287508
			NEXT HIGHER ASSEMBLY: -

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Wire closing improper		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Marking incomplete or incorrect		0.65%	3.2	Visual
203	Evidence of poor workmanship (see 3.8)		0.65%	3.2	Visual

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4.4.3 Testing

4.4.3.1 Moisture content of black powder (see 3.3) Major defect.-A sample of twenty-five black powder pellets shall be randomly selected from each eight hours production and shall be tested as specified in 4.5.1. Any failure to comply with the applicable requirements shall be cause for rejection of the eight hours production represented by the sample. If loading has begun, ignition cartridges loaded with the non-conforming black powder shall be rejected.

4.4.3.2 Weight of Propellant (see dwg. 9293422) weight above max, Major defect. Weight below minimum (min) Critical defect.-The propellant charge shall be weighed or volumetrically dispensed into a tared container and then check-weighed 100 percent. The initial weighing and the check-weighing shall be performed by different operators on different scales. Weight limits for check weighing should be established within the applicable weight requirement, depending on the accuracy and precision of the weighing system, to assure compliance with charge weight requirements. Any charge failing to meet the applicable requirement shall be considered defective and removed from the lot.

4.4.3.3 Weight of propellant in cartridge assembly (see dwg. 9293422) weight above max, Major defect. Weight below minimum (min), Critical defect.

4.4.3.3.1 315 sample cartridge assemblies shall be selected from each lot for this test. If four or more samples exceed the maximum charge weight or if any sample is below the minimum charge weight, the lot shall be rejected. Testing shall be as specified in 4.5.4.

4.4.3.3.2 Alternate procedure.-Loaded ignition cartridges shall be weighed 100 percent and the net weight of propellant determined by subtracting parts tare weight from the gross weight. The net weight of propellant should compare with the original weight of propellant within $\pm$ 0.5 grain.

4.4.3.4 Sensitivity

4.4.3.4.1 Non-functioning (see 3.4.1) Critical defect.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance

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criteria specified, 200 primed bodies, heads and firing plugs, as assembled, shall be randomly selected from each lot for this test. If one or more cartridges fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

4.4.3.4.1.1 After three consecutive lots have met the criteria of 4.4.3.4.1, 125 ignition cartridges shall be randomly selected from each lot for this test. If one or more ignition cartridges fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

4.4.3.4.2 Functioning (see 3.4.2) Major defect.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 500 primed bodies, heads and firing plugs, as assembled, shall be randomly selected from each lot for this test. If 4 or more primers fail to comply with the specified requirement, the lot shall be rejected. If 2 or 3 defectives are found a second sample of 500 primed bodies, heads and firing plugs, as assembled, shall be tested. If the combined number of defectives in both the first and second sample is 5 or more, the lot shall be rejected. Any primed body, head and firing plug assembly in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.2.2.

4.4.3.4.2.1 After three consecutive lots have met the criterial of 4.4.3.4.2, a sample of 315 primed bodies, heads and firing plugs, as assembled, shall be randomly selected from each lot for this test. If 3 or more primers fail to comply with the specified requirement, the lot shall be rejected. If 1 or 2 defectives are found a second sample of 315 primed bodies, heads and firing plugs, as assembled, shall be tested. If the combined number of defectives in both the first and second sample is 4 or more the lot shall be rejected. Any primed body, head and firing plug assembly in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.2.2.

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4.4.3.5 Leak Test (see 3.5) Major defect.-The ignition cartridges shall be tested 100 percent. Any ignition cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.3.

4.4.3.6 Velocity, Functioning and Security (see 3.6) Major defect.

4.4.3.6.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter N and an AQL of 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.84 exceeds the lot requirement, the lot shall be rejected. If one or more misfires occur (see para. 3.6.1) or if the metal parts security requirement is not met (see para. 3.6.2), the ignition cartridge lot shall be rejected. The test shall be performed as specified in para. 4.5.5.

4.4.3.6.2 After three consecutive lots have met the criteria of 4.4.3.6.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter K, and an AQL of 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.78 exceeds the applicable lot requirement, the lot shall be rejected. If one or more misfires occur (see para. 3.6.1) or if the metal parts security requirement is not met (see para. 3.6.2), the ignition cartridge lot shall be rejected. The test shall be performed as specified in para. 4.5.5.

4.4.3.7 Check test for deterioration of Primer, Percussion M35.-If the total elapsed time between original acceptance of the M35 primer lot and the assembly of that lot into the ignition cartridge exceeds two years, or if the M35 primers have been subjected to adverse conditions for any period of time since previous test, the M35 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46425 immediately before the M35 primer lot is assembled into the ignition cartridges.

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4.4.3.8 Compatibility of Tape at 90°C (see 3.7) Major defect.-A sample from every lot of tape supplied by one manufacturer shall be randomly selected and tested as specified in 4.5.7. If any sample fails to comply with the applicable requirement, the lot shall be rejected.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.3 (see 6.2).

4.5 Test Methods and Procedures

4.5.1 Moisture Content of Black Powder.-The moisture content of black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.5.2 Sensitivity

4.5.2.1 Non-functioning.-The primed body, head and firing plug, as assembled, shall be assembled to a suitable fixture. The firing pin of the test fixture shall be adjusted so that impact will be on the center of the firing plug of the primer. A weight of 2.95 pounds, minimum, shall be dropped on to the firing pin from a height of 3/16 inch, minimum, as if to function the primer. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which functions shall be classed defective. Metal parts used in this test shall not be returned to the lot.

4.5.2.2 Functioning.-The primed body, head and firing plug, as assembled, shall be assembled to an appropriate test fixture. The firing pin of the test fixture shall be adjusted so that impact will be 0.05 inch off center of the firing plug of the primer. A weight of 2.95 pound, maximum, shall be dropped on the firing pin to function the primer from a height of 1 3/4 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which fails to function shall be classed defective. Metal parts used in this test shall not be returned to the lot.

4.5.3 Leak Test.-The ignition cartridges shall be tested in accordance with drawing 9278622.

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4.5.4 Weight of Propellant.-The sample cartridge assemblies shall be cut open using a method that will not generate chips and all propellant removed and placed on the pan of a precision or analytical balance. The weight of propellant from each sample shall be determined to the nearest 0.01 grain.

4.5.5 Proving Ground Test.-The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The sample ignition cartridges shall be assembled to standard 81mm, HE, cartridges (weighed to 9.05 lbs) for which they were manufactured. The complete round, less propelling charges, shall be fired from a standard 81mm mortar at an elevation of forty-five degrees.

Reference rounds shall be fired without propelling charges for the purpose of velocity correction. At least one reference round should be fired for every three test rounds.

Individual velocities shall be recorded on the firing record for both reference and test rounds. Velocity standard deviations and arithmetic averages shall be caluclated and recorded. Temperature of the day and time of firing shall also be recorded In the event of a misfire, the misfired ignition cartridge shall not be disassembled by the proving ground but shall be forwarded to the responsible technical activity.

4.5.5.1 Refiring.-If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional cartridges as required shall be tested.

4.5.6 Propellant charge weight assessment.-Each lot of propellant shall be assessed. Ten assessment cartridges shall be loaded at charge weights of 110, 115 and 120 grains respectively, with a tolerance of plus or minus 0.5 grains. Cartridges shall be loaded using ignition cartridge components utilized by the production supplier. The assessment cartridges shall be assembled to complete rounds made up of reference components inert loaded to 9.05 pounds plus or minus .02 pounds. The rounds shall be conditioned at 70°F (± 2°F) for 24 hours prior to firing and shall be maintained at this temperature until fired. The rounds shall be fired from a

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standard 81mm mortar. Reference rounds shall be fired alternately with the three assessment loads for velocity correction. The mean corrected velocities of the three groups shall be used to establish the charge weight velocity relationship and the charge weight assessment shall be determined to the nearest grain. A charge assessment in excess of 115 grains shall not be used without prior approval of Picatinny Arsenal.

4.5.7 Determination of Compatibility of Tape at 90°C.- The reactivity value shall be determined in accordance with Method 403.1.2 of MIL-STD-286 with the following exceptions:

Paragraph 2.1-The specimens shall consist of 5.00 grams of propellant and 5.00 grams of tape.

Paragraph 4.1.2-After determining the volume of gas evolved from each heating tube, the reactivity value is calculated as follows:

$R = C - (A + B)$

A = amount of gas evolved by the propellant
B = amount of gas evolved by the tape
C = amount of gas evolved by the 50/50 mixture of propellant and tape.
R = reactivity value.

Paragraph 4.3-Place a weighed portion of 2.5 grams of the propellant in one heating tube, and 2.5 grams of tape in another heating tube. In a third heating tube place a 5 gram portion of a 50/50 mixture of the tape and propellant.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-Packaging shall be as specified on Drawings 9287507 and 9287505.

5.2 Packing and Marking.-Packing and marking shall be as specified on Drawings 9287506 and 9287508.

5.3 Transportation Regulations.-49 CFR 71-90 Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles shall be complied with.

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6. NOTES

6.1 Intended Use.-The XM299 ignition cartridge covered by this specification is intended for use on Cartridge, 81mm, HE, M374A3.

6.2 Ordering Data.-See MIL-A-48078 (PA).

6.3 Submission of Designs for Approval.-See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-A-A.

6.4 New Material Test Sample Requirements.-All requests for permission to use materials other than those specified on applicable drawings or specifications should be accompanied by a sample of the proposed substitute material to permit evaluation by Picatinny Arsenal. Prior to approval, all substitute materials must be tested for compatibility with M9 Propellant in addition to such testing required to evaluate the performance of the proposed substitute.

6.5 Submission of test data.-In addition to the normal distribution of records, when the ignition cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A
Dover, New Jersey 07801

6.6 Check test for deterioration of XM299 Ignition Cartridge.-If the total elasped time between original acceptance of the ignition cartridge lot and the assembly of that lot into the complete round exceeds two years, or if the ignition cartridges have been subjected to adverse conditions for any period of time since previous tests, the ignition cartridges shall be subjected to, and must satisfactorily comply with the following tests immediately before the ignition cartridges are assembled into the complete round.

6.6.1 Ignition cartridge assembly sensitivity

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6.6.1.1 Non-functioning.-The requirement shall be as specified in para. 3.4.1. The sampling plan for this test shall be as specified in para. 4.4.3.4.1. The test shall be performed as specified in 4.5.2.1.

6.6.1.2 Functioning.-The requirement shall be as specified in para. 3.4.2. The sampling plan for this test shall be as specified in para. 4.4.3.4.2. The test shall be performed as specified in 4.5.2.2.

6.5.2 Moisture content of black powder pellet.- Twenty-five sample black powder pellets shall be selected from the ignition cartridge lot. The requirement shall be as specified in para. 3.3. The test shall be performed as specified in para. 4.4.3.1.

Custodian:
ARMY-PA

Preparing Activity
ARMY-PA

Project Number: 1315-A-018

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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TELEPHONE NO.

DATE

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